Report 1: Epidemiological characterisation of the 2026 Bundibugyo virus disease outbreak in the Democratic Republic of the Congo

Prepared by the epidemiological intelligence team at INRB with national and international partners (U Oxford, Institut Pasteur). Contact tracing and line list data cutoff: October 9th, 2026; report prepared October 14th, 2026.

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Main findings

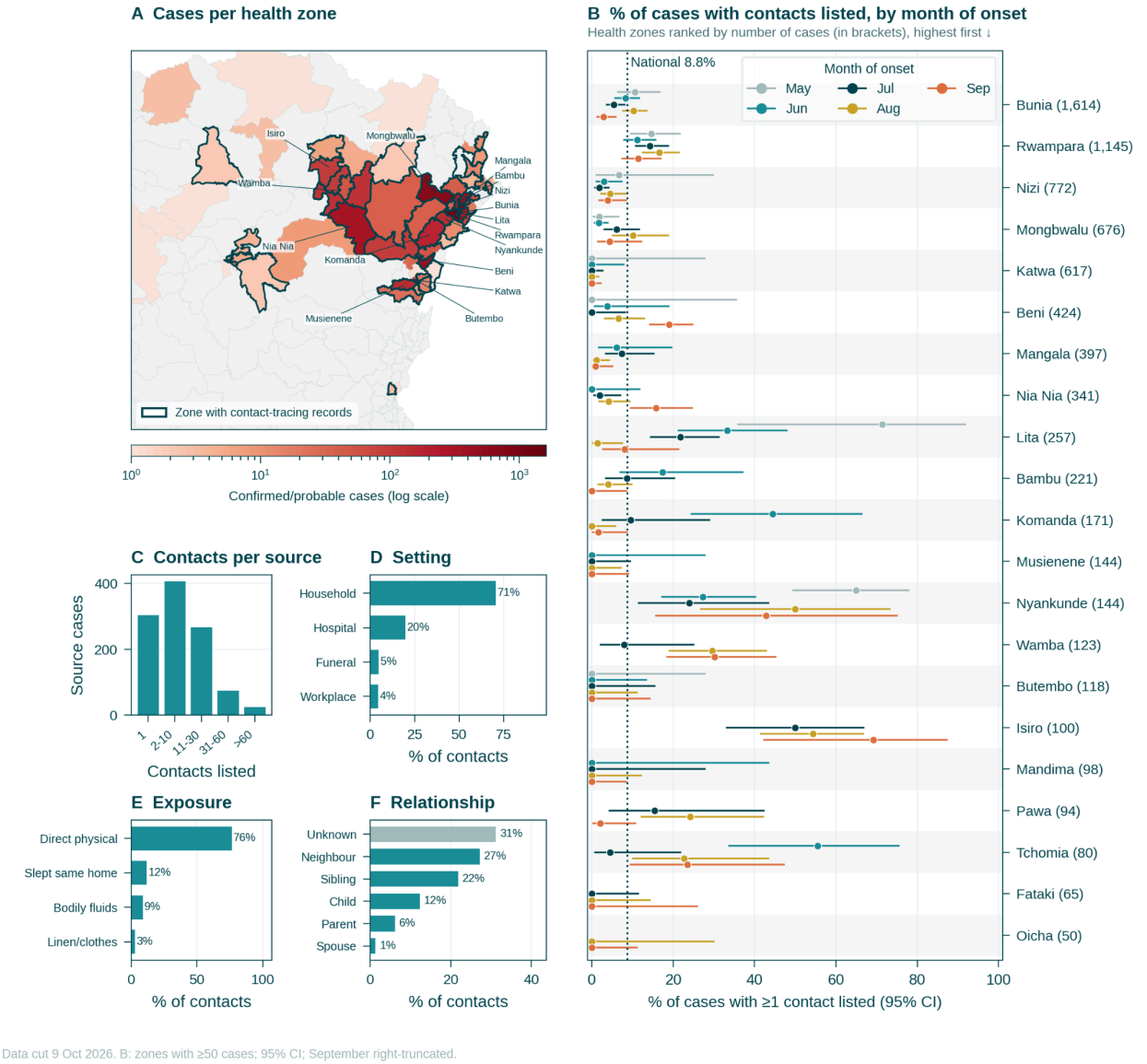
- Between the beginning of the outbreak and 9th October, 8.8% (717) of 8,145 confirmed/probable cases had contacts listed, and 45 (0.6%) new confirmed cases were already listed as contacts.
- Time from a source case's notification to first contact listed fell from a median of 7 days (June) to 2 days (September). However, due to delays between onset and notification, contacts of September-onset source cases are still listed a median of 6 days after the source case's symptom onset.
- Since 14 September, 74% of follow-up visits have a blank result (i.e. a follow-up visit was recorded, but the 'case status' of the contact was blank) vs 16% before. The median number of follow-up visits is 3 per individual, and only 26% of contacts are followed up ≥15 days[1] (21 days is the target).
- National secondary attack rate (SAR) over the entire study period was estimated to be 2.0% (95% CI: 1.4–2.8); highest for spouses (6.9%) and parents (3.9%) of confirmed cases, and hospital exposures (4.3%).
- The empirical serial interval (time interval between symptom onset of the infector and infectee) was estimated to have a median of 13.3 days (95% CI: 10.0-17.1). The incubation period (transmission-to-symptom-onset) was estimated to have a median of 8.6 days (95% CI: 6.5-11.0).
- 10% of cases contribute >80% of secondary cases. Most individuals (86%) do not have any detected onward transmissions.
- Katwa and Butembo (735 cases, 50% of Nord-Kivu's total) have no contact-tracing records in the DHIS2 database. In Beni, contact tracing only began in September, despite 73 BDBV-confirmed deaths already reported in the health zone by September 1st and an increasing case trajectory since mid-July.

Table 1: Key quantitative parameters.

Parameter	Overall	Key heterogeneities
Contact tracing coverage[2]	8.8% (717/8,145)	Among zones with any contact tracing and at least 20 source cases, Lolwa had the lowest coverage with 2.7%, Isiro the highest with 55%
Complete follow-ups[3]	24.2% (2,532/10,484)	Nord-Kivu had the lowest (1.9%), Ituri is highest at 28%. Some locations (e.g. Ariwara) reached follow up rates as high as 97%, but this is based on very few source cases.
Secondary attack rates (SAR)	2.0% (1.4-2.8%)	Haut-Uele has the highest documented SAR (3.5%, 1.8-5.5%); Bunia has 1.22% (0.56-2.06%).
Risk based on relationship (relative to unknown relationships)	-	Spousal and parent relationships had the highest SAR relative to the reference (2.80 and 1.59x reference, respectively)
Incubation period	Median 8.6 days (95% CI: 6.5-11.0, n=49)	-

[1]For contacts with a 'closed follow up window' i.e. were enlisted more than 21 days before the data cutoff of October 9th
[2]Proportion of cases registered as a source case with contacts listed on the database
[3]Proportion of contacts with a complete 'follow-up window' (follow-ups recorded up to 21 days post first follow-up)

Serial interval	Median 13.3 days (95% CI: 10.0-17.1, n=49))	Household settings may have a shorter serial interval than hospital settings by a median difference of 2.5 days (95% CI -6.0-9.5)
Time to first contact follow-up from source case notification	Median 2 days	-



Data cut 9 Oct 2026. B: zones with ≥50 cases; 95% CI; September right-truncated.

Figure 1. *Data summary. (A) Confirmed/probable cases per health zone. Health zones with a darker outline have source cases present in the contact tracing database. (B) Share of cases that were listed as source cases (had ≥1 contact listed) over time in all health zones where contact tracing has been taking place, in comparison to the national mean over the outbreak. (C) Distribution of number of contacts listed per source case. (D–F) Setting, exposure type and relationship for all source–contact pairs.*

2. Data summary

The contact-tracing database lists 1,073 unique source persons and 12,846 contacts (13,143 source–contact pairs). 717 (66.8%) source persons are confirmed/probable cases in the linelist; 181 could

not be matched between the two databases and 175 are classified as not a case, blank or suspected only. Source persons named a median of 6 contacts (IQR 1–15; mean 12.2; n=1,073; Table 2); 28% named only one, while 44 named ≥50. Contacts between source cases and contacts were often in the household (70.7%) or hospital (19.7%), mostly through direct physical contact (76.4%; Figure 1).

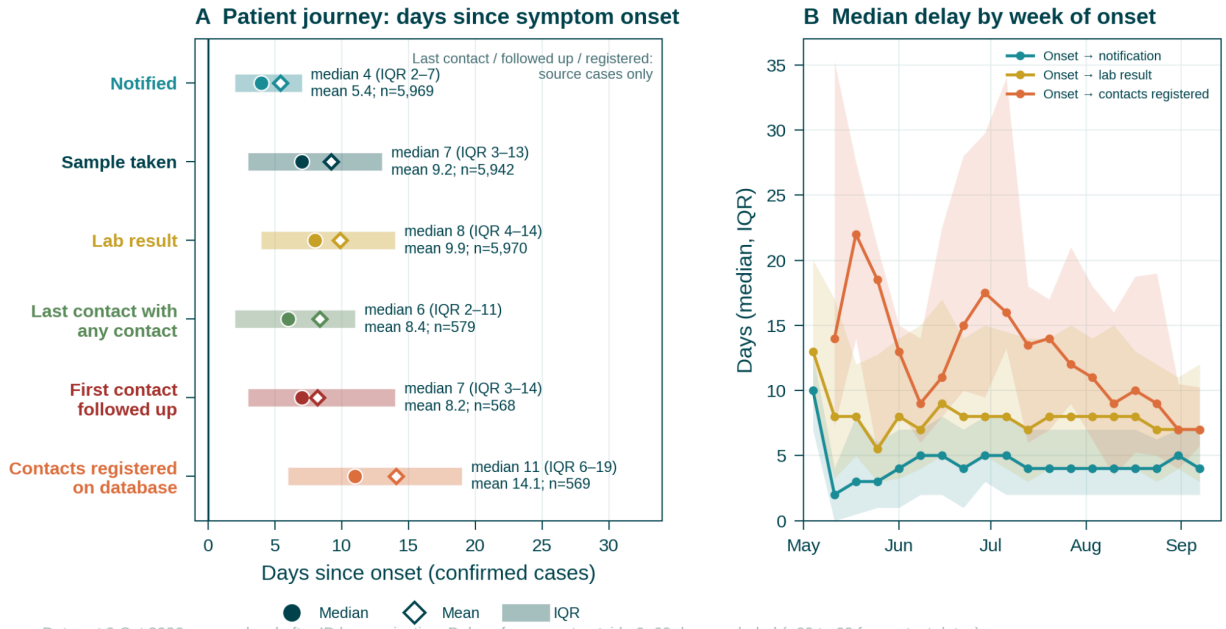
Table 2. Contact-tracing indicators by province (province of source case for CT columns).

Province	Cases	Deaths	% cases with contacts listed	% cases known contact before notif.	Source cases*	Contacts	Median contacts / source	% contacts fully followed up	SAR % (k/n)
Ituri	6,247	2,253	8.6	0.5	683	8,574	7	28	2.1 (144/6,943)
Nord-Kivu	1,452	646	3.9	0.6	67	2,378	17	2	0.8 (8/1,059)
Haut-Uele	**373**	**57**	**30.3**	**1.9**	**162**	**1,416**	**6**	**19**	**3.5 (26/753)**
Tshopo	54	8	13.0	0.0	10	77	5	12	1.6 (1/64)
Bas-Uele	15	1	13.3	0.0	2	82	41	67	0.0 (0/82)
Sud-Kivu	4	1	25.0	0.0	2	34	17	18	0.0 (0/10)
National	8,145	2,966	8.8	0.6	1,073	12,846	6	24	2.0 (179/8,911)

**National totals include 36 source persons with Kasai (Bulape) alert-ID prefixes, 109 with unlocatable IDs and 2 in Tshuapa.*

3. Patient journey

Among confirmed cases with onset prior to September 13th, it took a median of 4 days (IQR 2–7; mean 5.4; n=5,969) after symptom onset for cases to be notified; a further 3 days until a sample was taken (median 7 days (IQR 3–13; n=5,942) after symptom onset) but lab results were often returned within the same day, or one day later (IQR 0-1, Figure 2A). For cases that became source cases, their first contact was followed up a median of 3 days (IQR 1–7; mean 8.2 from onset; n=568) after source case notification. Median delays from onset to notification and lab result have remained relatively stable over time, but time from source case onset to contact registration has been decreasing since July - this may be due to right truncation, however (Figure 2B).



Data cut 9 Oct 2026; person-level after ID harmonisation. Delays from onset outside 0–60 days excluded (−60 to 60 for contact dates).
A: onset on or before 13 Sep (≥26 days to data cut); last contact = latest last-exposure date among the source case's contacts.
B: weeks with ≥5 cases; onset weeks to 7–13 Sep only (≥26 days follow-up to data cut).

Figure 2. *Patient journey of confirmed cases. (A) Median, mean and IQR of days from symptom onset to notification, sample collection, lab result, last contact with any contact, first follow-up of a contact (source cases only), and first contacts registered on the database (source cases only). (B) Weekly median and IQR (shaded) of selected delays, by week of onset.*

4. Effectiveness of contact tracing

The weekly share of new cases that became source cases has stayed between about 5% and 15% since June (Figure 3B). Newly enrolled source cases rose to a peak of approximately 92 per week in mid-September, when 1,941 new contacts were listed in one week, mostly from Beni. However, follow-up on these newly listed contacts has been low (Figure 3C, 3D). Only ~0.6% of confirmed cases entered into the line list were previously listed as contacts of confirmed cases (Figure 3A, orange line). First contact follow-up was on average 2 days (IQR 0–5; n=695) after source case notification (not confirmation). We observe an improvement over time in the delay between source case symptom onset and registration of its contacts (Figure 2B, orange line). A median of 9 follow-up visits were recorded per contact, but only 3 (IQR 1–15) days on average had a non-blank entry for contact status (whether the contact had become a case, a suspect case, or was still not a case); 26% had ≥15 days with a result. Completeness differs between areas: Nord-Kivu and Haut-Uele contacts have almost no recorded results (Table 2, Figure S3). The health zone of Nyankunde has performed well relative to other health zones, in terms of cases listed as source cases for contact tracing (Figure 1B), and Komanda and Lita outperform other health zones in terms of proportion of contacts with full follow-up records from May to July; however, Komanda has not listed a significant number of contacts in recent months, and Lita did not list a significant number of contacts in August (Figure 3D).

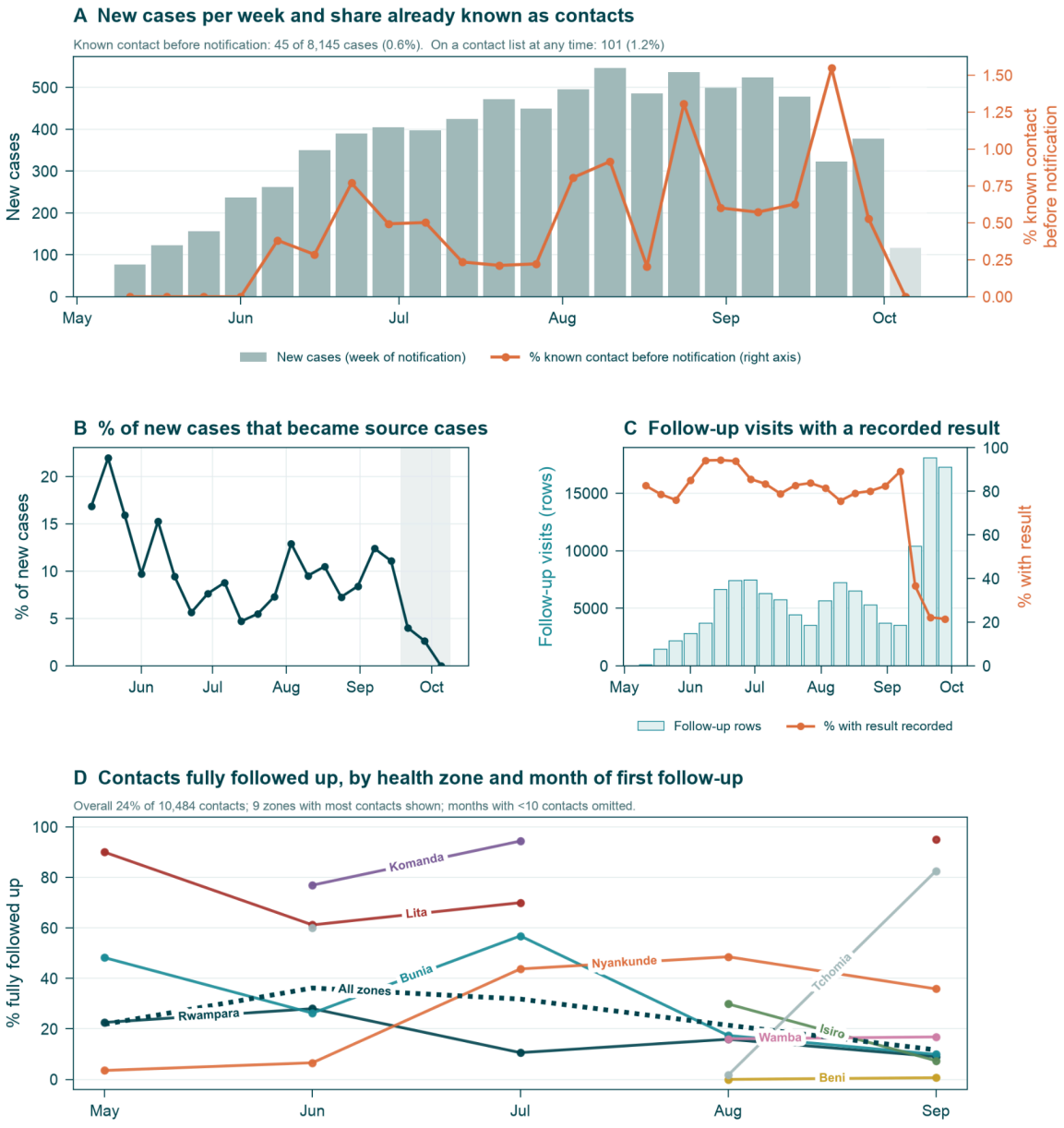


Figure 3. Effectiveness of contact tracing. (A) New cases by week of notification (bars) and proportion of new cases that had previously been identified as contacts (line). (B) Percentage of new cases that became source cases, i.e. had their contacts registered to the contact tracing database. (C) Follow-up visit rows per week and share with a result recorded. (D) Proportion of contacts 'fully followed up', i.e. with follow-up results recorded up to 21 days post their most recent exposure to their source case. The 9 health zones with the highest number of contacts are shown against the national average (dotted line).

5. Epidemiological implications

Of 8,911 contacts of confirmed/probable sources with a closed follow-up window[4], 179 became confirmed or validated cases according to either the linelist or the contact tracing follow-ups, giving a crude Secondary Attack Rate (SAR) of 2.0% (95% CI 1.4–2.8). This includes 64 'case validated' entries in the contact tracing that are not found in the linelist, the majority (n=59) of which originate from hospital settings in Nyankunde from May 17th to June 2nd (Figure S4, S5). Excluding these cases reduces the national SAR to 1.3% (1.0-1.7%).

SAR was slightly higher for spouses (RR 2.80, 0.89–7.22) and parents (RR 1.59, 0.75–3.68) than for contacts with 'unknown' relationship (Figure 4A), although CIs included 1. Hospital exposures were associated with a higher SAR compared to household exposures (RR 3.34, 1.45-6.11). This effect remained after mutual adjustment by fitting a single logistic regression that includes exposure type, setting, relationship, source-case province, source outcome and month of listing as covariates (SI-5). Haut-Uele had a higher estimated SAR (3.5%, 95% CI 1.80-5.50%), possibly because more of its cases were traced (30% of 373 cases compared to only 8.6% of 6,247 traced in Ituri province).

The serial interval was estimated to have a median of 13.3 days (95% CI 10.0–17.1; n=49 eligible pairs according to our inclusion criteria in SI-1; Figure 4B). Household settings (median 11.0, 95% CI 9.0-17.0) report a shorter serial interval than hospital settings (median 13.5, 95% CI 9.5-21.5; Figure 4C). However these estimates are based on few infector-infectee pairs (n=29,18 for household, hospital) and do not account for differences in 'exposure opportunity times' between settings - i.e. a case that is hospitalised can no longer transmit in the household going forward. Both are lower than for EVD[1]. Time from possible transmission to onset was a median 8.6 days (95% CI 6.5–11.0; n=43, Figure 4D), where the possible transmission window was taken as the interval between source case symptom onset and the date of last exposure to the source case. The incubation period estimate is similar to that estimated for EVD[1].

The secondary case offspring distribution is highly heterogeneous. We find that 10% of cases contribute over 80% of cases (negative binomial offspring distribution parameter k = 0.12, 95% CI 0.09 - 0.17, 642 source cases and 179 secondary cases, Figure S9). When excluding the aforementioned 'case validated' entries (mainly concentrated in Nyankunde) k increases to 0.25 (Figure S9). These parameters are in range of previous Ebola outbreaks[1].

[4] First follow up date on or before 21 days prior to the data cutoff of 9th October

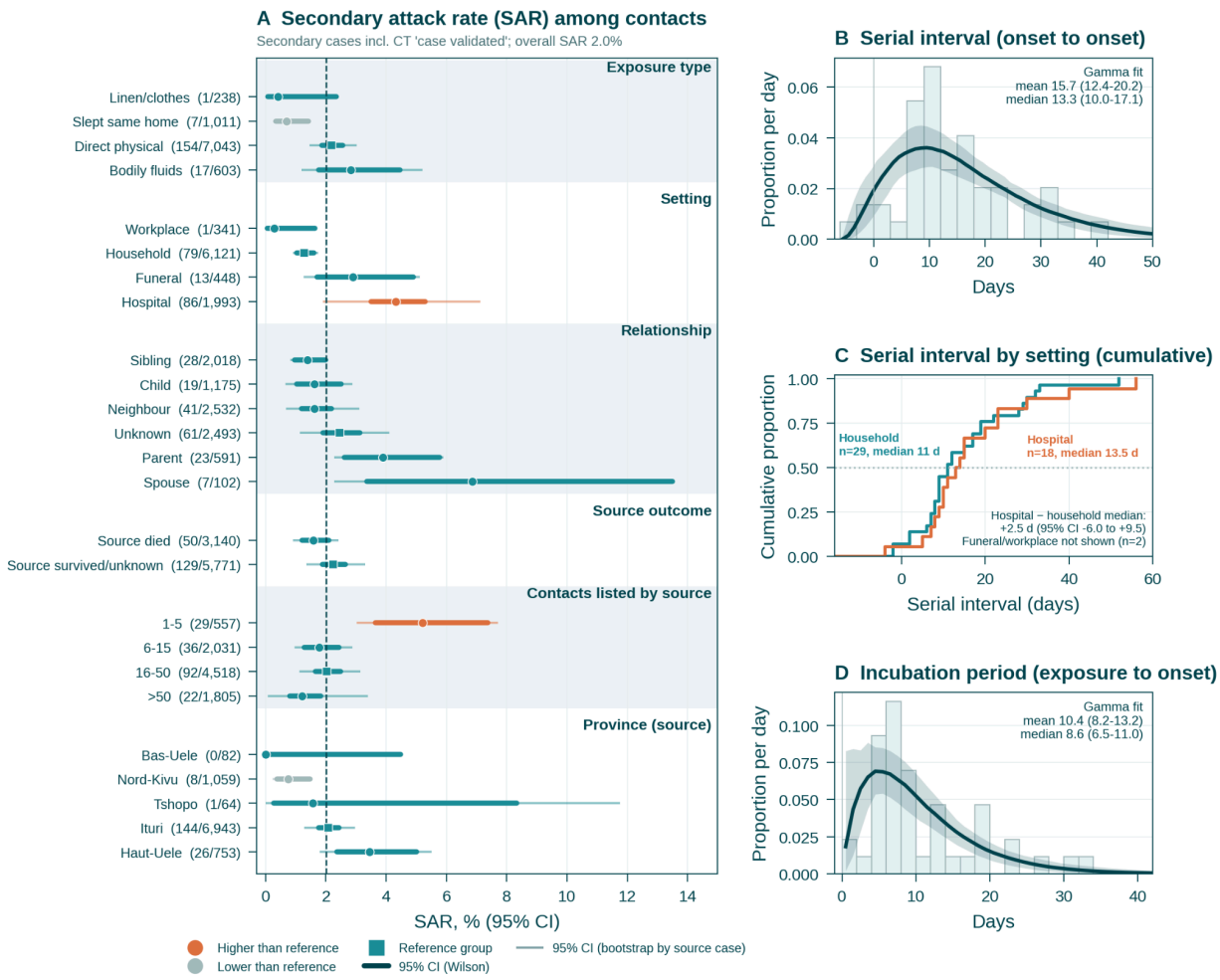


Figure 4. *Epidemiological implications. (A) SAR (95% Wilson CI) by exposure type, setting, relationship, source outcome, number of contacts named by the source and province (levels with ≥50 contacts). Coral = RR CI above 1; grey = below 1. (B) Serial interval, eligible infector-infectee pairs. (C) Cumulative distribution of serial interval, stratified by exposure setting for household and hospital settings (insufficient data for funeral and workplace settings) (D) Days from middle of exposure window (see definitions) to onset in secondary cases for eligible infector-infectee pairs.*

6. Limitations

While steps were taken to account for truncation in recent data by filtering to source case onsets prior to August 28th in key analyses, some metrics (e.g. delay from onset to contact registration in Figure 2B) may still be impacted by right-truncation and interval censoring for delays, and generally incomplete data. Contrastingly, many secondary cases were linked to a source only after their own notification. SAR and risk-factor estimates may therefore be inflated by retrospective listing of contacts who were already cases (particularly for the Nyankunde hospital cluster, which demonstrated very high secondary attack rates; Figure S5).

Linkage between the linelist and contact tracing data depends on probabilistic name/demographic matching; missed links will lower the SAR and the 'known contact' share. Serial interval and incubation estimates use

few pairs (n=49, n=43 respectively) from a subset of pairs where key dates are reported. There have been persistent challenges with reporting of key dates (e.g., symptom onset) and this will impact our analysis.

Analysis code to reproduce the analysis presented here is at: https://github.com/INRB-UMIE/BDBV2026-Contact_Tracing_Analytics

Technical appendix

SI-1 Inclusion Criteria for Main Report Analyses

Metric	Data	Description
Secondary Attack Rate	Source Case	Source cases were included if they were found in the linelist and had a recorded status of 'confirmed case' or 'probable case'. If duplicate entries were present in the linelist, the status was decided according to the following ranking: *confirmed > probable > suspected > not_a_case > blank*
	Contact of Source Case	Contacts (denominator for SAR) were included if they were listed as a contact for a confirmed/probable source case (above), and if their follow up window is closed (i.e. their first follow up date was on or before 18th September, 21 days prior to the data cutoff of 9th October)
	Positive Contact	Contacts were considered to have become a positive case if they had a status of 'confirmed case' or 'probable case' in the linelist, or if they had a status of 'case confirmed' or 'case validated' according to the contact tracing follow-up data. Linelist status takes precedence over contact-tracing, i.e. if the linelist records a contact as 'not a case', that individual is not counted, even if they were a 'case confirmed' according to the contact tracing data.
Serial Interval	Infector-Infectee Pairs	Infector-Infectee pairs to fit serial interval (delay between infector & infectee symptom onset) distributions were selected using all source case-contact pairs where members fit the below criteria for each role. Bi-directional pairs (pairs where two individuals were listed as contacts with each other as their source cases) were excluded.
	Infector	Infectors were eligible if: (1) they were recorded as a confirmed or probable case according to their matched linelist entry, and (2) their symptom onset was non-blank and on or before August 28th (42 days prior to the data cutoff of 9th October)
	Infectee	Infectees were eligible if: (1) their source case was an eligible infector, and (2) they were recorded as confirmed or probable case according to their matched linelist entry, and (3) their symptom onset date was non-blank and between -5 and 60 days (inclusive) of their infector's symptom onset
Incubation Period	Infector-Infectee Pairs	Infector-Infectee pairs to fit incubation period (time from exposure to symptom onset) distributions were selected using all source case-contact pairs where members fit the below criteria for each role. Bi-directional pairs (pairs where two individuals were listed as contacts with each other as their source cases) were excluded.
	Infector	Infectors were eligible if: (1) they were recorded as a confirmed or probable case according to their matched linelist entry, and (2) their symptom onset was non-blank and on or before August 28th (42 days prior to the data cutoff of 9th October)
	Infectee	Infectees were eligible if: (1) their source case was an eligible infector, and (2) they were recorded as confirmed or probable case according to their matched linelist entry, and (3) their 'last contact date' with their infector was non-blank, and (4) their symptom onset date was: (a) non-blank, and (b) after their infector's symptom onset, and

		(c) no more than 60 days after infector's symptom onset, and (d) no more than 42 days after their last contact with their infector

SI-2 Definitions

Source case: Any person appearing as source_case_person_id in the CT data, whatever their linelist status; process indicators use all of them.

Secondary case: A contact who is a confirmed/probable linelist case, OR whose CT follow-up result was ever 'case_confirmed' or 'case_validated', unless the linelist classifies them as not a case.

Known contact before notification: Case whose person_id is a CT contact with first registration on or before their earliest linelist notification date.

Follow-up window: 21 days from first contact follow up. A closed window (first follow up on or before 18th of September) is defined as 'complete' if a result is recorded on or after 20 days from the first follow-up date, or if the contact became positive.

Exposure window: Time between infector symptom onset and date of last exposure to source case for an eligible infector-infectee pair according to inclusion criteria in SI-1.

SAR denominator: Pairs whose source is a confirmed/probable case and whose 21-day window was closed by 9 October (first follow up on or before 18th of September).

Serial interval: onset(contact) – onset(source) for pairs included according to the inclusion criteria in SI-1.

Incubation period: onset – exposure window, for linelist-linked secondary cases, for pairs included according to the inclusion criteria in SI-1.

SI-3. Harmonisation of duplicates and discrepancies

Decisions taken while merging duplicates or fixing inconsistent values, with the number of records or persons affected. 'Primary record' = the person's linelist row with the highest classification, then the most non-missing fields.

Dataset	Issue	N	Decision
linelist	rows in linelist export	58,828	all alerts (cases, non-cases, unclassified)
linelist	distinct person_id	58,446	unit of analysis = person_id
linelist	health_zone 'Bili' (Bas-Uele)	71	renamed 'Bili (Bas-Uele)' to match shapefile
linelist	date_of_symptom_onset: implausible date (<2026-04-01 or >2026-10-09)	110	set to missing
linelist	date_of_notification: implausible date (<2026-04-01 or >2026-10-09)	57	set to missing
linelist	investigation_datetime: implausible date (<2026-04-01 or >2026-10-09)	77	set to missing
linelist	date_of_sample_collection: implausible date (<2026-04-01 or >2026-10-09)	2	set to missing
linelist	lab_analysis_date: implausible date (<2026-04-01 or >2026-10-09)	1	set to missing
linelist	persons with >1 linelist row	366	collapsed to one row per person (rules below)
linelist	multi-row persons including an 'undecided' ID match	12	kept as one person (person_id taken as given)
linelist	conflicting final classification	216	highest-ranked kept: confirmed > probable > suspected > not_a_case > blank (a positive lab result is treated as definitive)
linelist	conflicting date_of_symptom_onset	130	earliest plausible date kept (first-reported symptoms)
linelist	conflicting date_of_notification	130	earliest kept (first time health authorities were notified)

linelist	conflicting date_of_sample_collection	68	taken from the primary record (highest classification, then most complete); earliest if primary missing
linelist	conflicting lab_analysis_date	57	taken from the primary record; earliest if primary missing
linelist	conflicting sex	37	primary record value; missing filled from other rows
linelist	conflicting age	90	primary record value; missing filled from other rows
linelist	conflicting health_zone	20	primary record value (zone of the confirmed/primary alert)
linelist	conflicting type_of_alert	25	person counted as died if ANY row is a death alert or sample status 'deceased'
linelist	cases (confirmed + probable persons)	8,145	case definition used throughout; suspected (n=182) excluded
linelist	lab analysis date before sample date	108	lab analysis date set to missing
linelist	cases with no onset date	1,238	date of notification used for epi curve timing
CT	rows in combined CT file (all exports, incl. aged-out contacts)	132,948	combined file used (processed file holds only contacts current on 9 Oct)
CT	case_circumstances raw value '2.Hôpital'	3	recoded to 'hospital'
CT	case_circumstances raw value '1.Foyer'	1	recoded to 'household'
CT	relationship_to_case raw value 'Inconnue'	2	recoded to 'unknown' (code not in data dictionary)
CT	relationship_to_case raw value 'Epoux/ Épouse'	2	recoded to 'spouse'
CT	contact_type raw value '1.Contact physique direct'	4	recoded to 'direct_physical_contact'
CT	contact_type raw value 'MVE-005'	139	recoded to 'unknown' (code not in data dictionary)
CT	contact_type raw value '1'	74	recoded to 'unknown' (code not in data dictionary)
CT	contact_type raw value '2'	11	recoded to 'unknown' (code not in data dictionary)
CT	followup_result raw value 'Non Cas'	2	recoded to 'not_a_case'
CT	followup_result raw value 'Confirmé'	1	recoded to 'case_confirmed'
CT	followup_result raw value 'DB'	10	recoded to 'other_result' (code not in data dictionary)
CT	case_circumstances missing	607	set to 'unknown'
CT	relationship_to_case missing	592	set to 'unknown'
CT	contact_type missing	20	set to 'unknown'
CT	last_contact_with_source_date: implausible date (<2026-04-01 or >2026-10-09)	49	set to missing
CT	followup_date: implausible date (<2026-04-01 or >2026-10-09)	53	set to missing
CT	rows where source person = contact person (self-pairs)	38	dropped (ID-matching error)
CT	duplicate follow-up rows (same pair, date, result; e.g. same contact under 2 IDs)	2,248	dropped
CT	pairs recorded in both directions (A->B and B->A)	1	kept; excluded from serial-interval estimate
CT	pairs with conflicting contact_type	54	most frequent non-unknown value kept; ties -> value in most recent export
CT	pairs with conflicting relationship_to_case	41	most frequent non-unknown value kept; ties -> value in most recent export
CT	pairs with conflicting case_circumstances	43	most frequent non-unknown value kept; ties -> value in most recent export
CT	pairs with conflicting last-contact date	30	latest date kept (follow-up clock runs from last exposure)
CT	pairs with conflicting registration date	110	earliest kept
CT	pairs with >1 contact_id (same person registered twice)	160	merged into one pair via contact_person_id
CT	pairs whose follow-up results differ across visits	247	most severe result retained (confirmed > validated > suspected > other/recycled > not a case)
CT	source persons not found in linelist	181	kept as source cases; province/health zone inferred from alert-ID prefix (280 pairs still unlocated)
CT	contacts confirmed/probable in linelist	109	secondary case
CT	contacts confirmed/probable in linelist but CT outcome never updated (worst CT result 'not a case' or blank)	62	linelist classification wins (secondary case)
CT	contacts with CT result 'case_confirmed' but no linked linelist case	57	counted as secondary case (CT team reports confirmation; linkage to linelist missing)

CT	contacts with CT result 'case_confirmed'/'case_validated' but linelist says not_a_case	5	NOT counted as secondary case (lab-based linelist classification wins)
CT	contacts with CT result 'case_validated' only	103	Counted in main analysis as secondary case; investigated in in sensitivity analyses (Figure S5, S9)
CT	source persons classified not_a_case in linelist	75	kept in CT process metrics; excluded from SAR/serial interval

SI-4. Delays by province and month

Level	Group	Onset→notification	Notification→sample	Sample→lab result	Onset→lab result	Notification→first contacts registered	Onset→first contacts registered
Month	2026-05	3 (1–7)	0 (0–3.5)	1 (0–4)	8 (4–14)	16 (1.5–25)	19 (8–27)
Month	2026-06	4 (2–7)	1 (0–6)	0 (0–1)	8 (4–15)	7 (3–14)	11.5 (8–19.75)
Month	2026-07	4 (2–7)	1 (0–6)	0 (0–1)	8 (4–15)	6.5 (2–13)	14.5 (8–21)
Month	2026-08	4 (2–7)	1 (0–5)	0 (0–1)	8 (4–14)	3 (1–9)	10 (5–18)
Month	2026-09	4 (2–7)	1 (0–3)	0 (0–1)	6 (3–10)	2 (1–4)	6 (4–10)
Month	2026-10	2 (0–4)	0 (0–1)	1 (0–1)	3 (2–5)	–	–
Province	Bas-Uele	–	0 (0–0.5)	1 (0–2)	–	–	–
Province	Haut-Uele	4 (2–7)	1 (0–9)	1 (0–2)	9 (5–15)	3 (1–12.5)	9.5 (5–18)
Province	Ituri	4 (2–7)	1 (0–7)	0 (0–1)	8 (4–14)	5 (1–13)	11 (6–19)
Province	Nord-Kivu	3 (1–6)	0 (0–0)	0 (0–1)	4 (2–7)	3 (2–4)	9 (6–12)
Province	Tshopo	5 (2–8.5)	0 (0–1)	0 (0–0)	6.5 (3–9)	–	–

SI-5. Risk factors for becoming a secondary case

Denominator: 8,911 source–contact pairs (contacts of confirmed/probable source cases whose first follow-up was on or before 18 Sep 2026). Secondary case: confirmed/probable in the linelist or CT 'case confirmed'/'case validated'. Reference level: 'unknown' for relationship; otherwise the most common level. Adjusted OR: single logistic regression including exposure type, setting, relationship, source province, source outcome and month listed; levels with no secondary cases are dropped ('–'). 95% CI (bootstrap): 2,000 resamples of source cases, accounting for clustering of contacts within source cases (used in the text); Wilson / Wald CIs assume independent contacts. Bootstrap CIs are unreliable for levels with fewer than 5 secondary cases (shown in grey) and are 0.0–0.0 where there were none.

Factor	Level	k/n	SAR %	SAR 95% CI (bootstrap)	SAR 95% CI (Wilson)	Crude RR	RR 95% CI (bootstrap)	RR 95% CI (Wald)	Adjusted OR	aOR 95% CI (bootstrap)	aOR 95% CI (Wald)
Exposure type	direct physical contact	154/7,043	2.2	1.5–3.0	1.9–2.6	ref			ref		
	sleeping in the same home	7/1,011	0.7	0.2–1.3	0.3–1.4	0.32	0.11–0.66	0.15–0.67	0.43	0.14–0.95	0.19–0.93
	contact with bodily fluids	17/603	2.8	1.2–5.2	1.8–4.5	1.29	0.60–2.23	0.79–2.11	1.29	0.55–2.59	0.75–2.22
	handling contaminated clothes or linen	1/238	0.4	0.0–1.5	0.1–2.3	0.19	0.00–0.70	0.03–1.37	0.17	0.00–0.74	0.02–1.23
	unknown	0/16	0.0	0.0–0.0	0.0–19.4	–			–		
Setting	household	79/6,121	1.3	0.9–1.7	1.0–1.6	ref			ref		
	hospital	86/1,993	4.3	1.9–7.1	3.5–5.3	3.34	1.45–6.11	2.47–4.52	3.03	1.52–5.21	2.09–4.38
	funeral	13/448	2.9	1.3–5.1	1.7–4.9	2.25	0.94–4.46	1.26–4.01	2.22	0.89–5.18	1.18–4.18
	workplace	1/341	0.3	0.0–0.9	0.1–1.6	0.23	0.00–0.76	0.03–1.63	0.27	0.00–1.07	0.04–1.98
	unknown	0/8	0.0	0.0–0.0	0.0–32.4	–			–		
Relationship	neighbour	41/2,532	1.6	0.7–3.1	1.2–2.2	0.66	0.29–1.47	0.45–0.98	0.72	0.27–1.70	0.47–1.10
	unknown	61/2,493	2.4	1.1–4.1	1.9–3.1	ref			ref		
	sibling	28/2,018	1.4	0.8–2.1	1.0–2.0	0.57	0.27–1.31	0.36–0.88	0.68	0.31–1.46	0.42–1.11
	child	19/1,175	1.6	0.7–2.9	1.0–2.5	0.66	0.24–1.64	0.40–1.10	0.89	0.30–2.13	0.51–1.58
	parent	23/591	3.9	2.3–5.9	2.6–5.8	1.59	0.75–3.68	0.99–2.55	2.64	1.20–5.85	1.51–4.60
	spouse	7/102	6.9	2.3–12.0	3.4–13.5	2.80	0.89–7.22	1.32–5.98	4.25	1.19–10.98	1.80–10.03
Province (source)	Ituri	144/6,943	2.1	1.3–3.0	1.8–2.4	ref			ref		
	Nord-Kivu	8/1,059	0.8	0.2–1.3	0.4–1.5	0.36	0.11–0.76	0.18–0.74	0.75	0.18–2.94	0.32–1.74
	Haut-Uele	26/753	3.5	1.8–5.5	2.4–5.0	1.66	0.81–3.31	1.10–2.51	2.01	0.70–7.46	1.16–3.48
	Bas-Uele	0/82	0.0	0.0–0.0	0.0–4.5	–			–		
	Tshopo	1/64	1.6	0.0–11.8	0.3–8.3	0.75	0.00–6.06	0.11–5.30	0.73	0.00–9.14	0.10–5.46
	Sud-Kivu	0/10	0.0	0.0–0.0	0.0–27.8	–			–		
Month listed	2026-05	2/118	1.7	0.0–6.7	0.5–6.0	1.00	0.00–4.41	0.25–4.09	0.89	0.00–4.20	0.21–3.84
	2026-06	85/2,207	3.9	1.7–6.6	3.1–4.7	2.28	0.93–4.73	1.60–3.25	2.51	1.11–4.83	1.67–3.77
	2026-07	14/1,433	1.0	0.4–1.7	0.6–1.6	0.58	0.24–1.17	0.32–1.05	0.63	0.26–1.36	0.34–1.18
	2026-08	46/2,727	1.7	1.1–2.4	1.3–2.2	ref			ref		
	2026-09	31/2,373	1.3	0.7–2.3	0.9–1.8	0.77	0.36–1.59	0.49–1.22	0.87	0.25–2.49	0.52–1.46

	2026-10	1/53	1.9	0.0–9.1	0.3–9.9	1.12	0.00–5.46	0.16–7.96	0.90	0.00–7.56	0.11–7.07
Source outcome	Source survived/unknown	129/5,771	2.2	1.4–3.3	1.9–2.6	ref			ref		
	Source died	50/3,140	1.6	0.9–2.4	1.2–2.1	0.71	0.36–1.36	0.52–0.98	0.91	0.47–1.72	0.64–1.30
Contacts listed by source	1-5	29/557	5.2	3.0–7.7	3.6–7.4	2.56	1.29–5.25	1.70–3.85	-		
	6-15	36/2,031	1.8	1.0–2.9	1.3–2.4	0.87	0.41–1.82	0.59–1.28	-		
	16-50	92/4,518	2.0	1.1–3.2	1.7–2.5	ref			-		
	>50	22/1,805	1.2	0.1–3.4	0.8–1.8	0.60	0.03–1.98	0.38–0.95	-		

SI-6. Fitting distributions for serial interval and incubation period

We fit delay distributions for the serial interval and incubation periods using a Bayesian framework (Epidist [2] in R, with the exception of brms for negative binomial fits).

For both delay types, we tested multiple distribution families, including Negative Binomial, Lognormal, Gamma, and Weibull, and selected the best-fit distribution using the leave-one-out cross-validation information criterion (LOOIC, Figure S6). Weibull distributions consistently failed to converge, so are not shown here. Gamma was best across most scenarios and assumptions and results from the Gamma fits are shown in the main report text.

For serial interval models shown in the main report text, negative delays up to -5 days in the serial interval were retained (accounting for heterogeneities in time from exposure to symptom onset). Where the data included negative delays, we could fit a normal distribution directly to the data, but for the other families it was necessary to transform the data forward (+6 days), before transforming the resulting fits backwards. In Figure S7 we test the sensitivity to this assumption by fitting distributions to exclusively positive serial interval data. In accounting for right truncation for the estimates shown in the main report text, we cut the data to only include infector-infectee pairs where the symptom onset date of the infector fell on or before August 28th (sensitivity to this decision also shown in Figure S7).

For the incubation period (time from infection to symptom onset) we similarly tested the sensitivity of the model fit to the infector onset date cutoff of August 28th (Figure S8). We assumed that transmission was possible between symptom onset of the infector and date of last exposure between infectee and infector using a uniform distribution (not all interactions were recorded, only the last exposure). We also tested for sensitivity of these estimates to the alternative assumption that transmission took place on the date of last exposure (Figure S8). In future work it will be important to characterise how infectiousness can vary since recorded symptom onset, e.g., by using Ct values.

SI-7. Supplementary figures



Figure S1. Weekly new cases (grey), cases who became source cases (dark teal) and new contacts listed (orange, right axis) in the six zones with most cases.

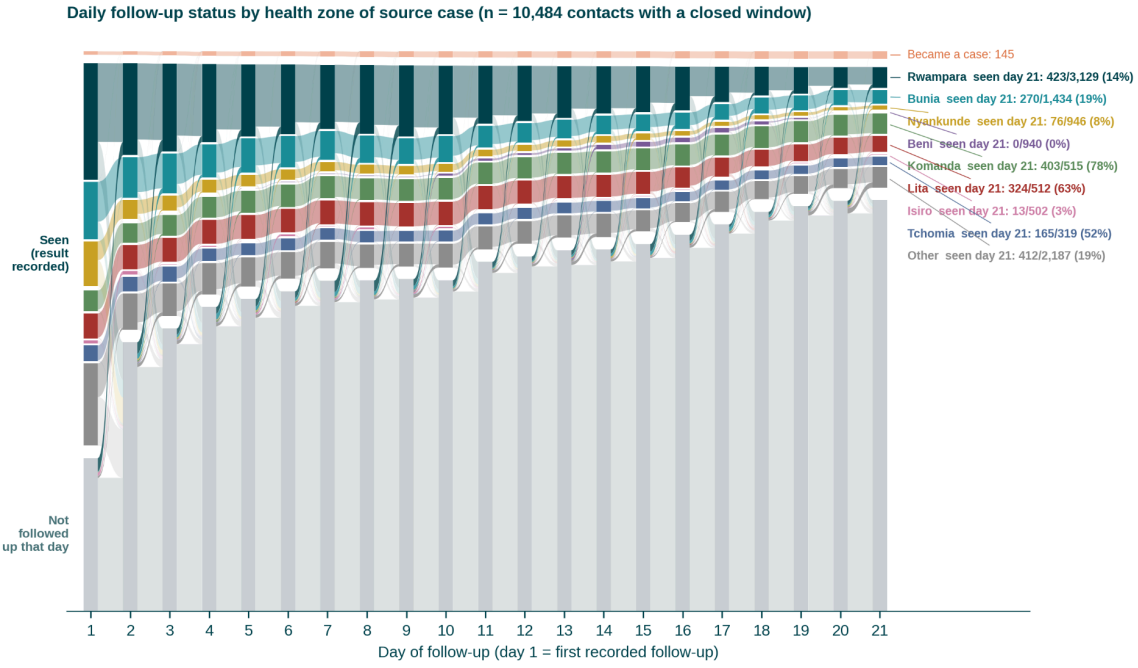
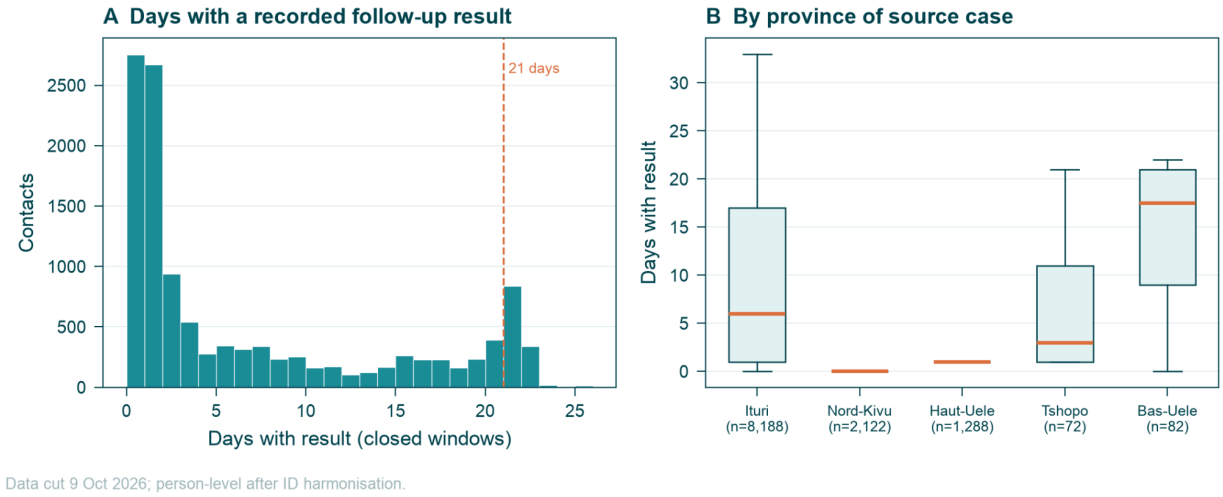
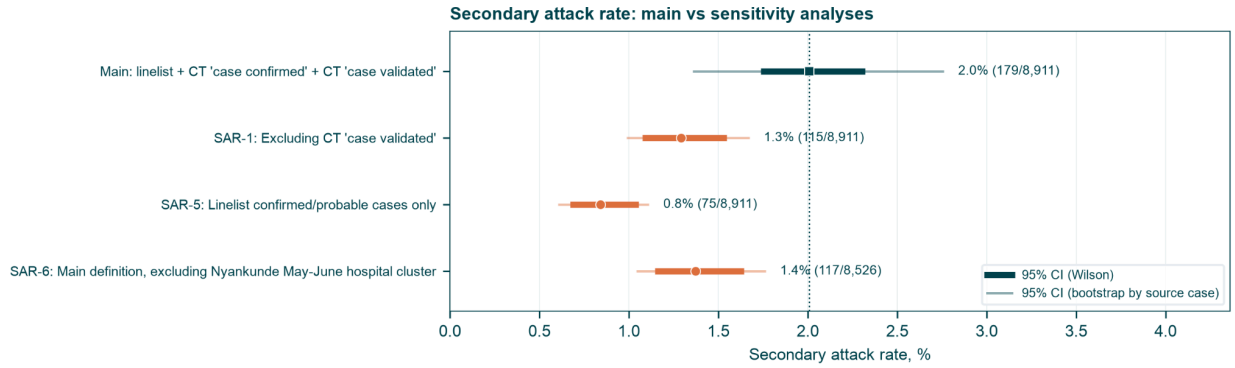


Figure S2. Sankey plot of contact follow-ups from day of first follow up to the end of the target follow up window (21 days). Each column shows contacts seen that day (a follow-up result recorded; coloured by health zone), not followed up that day (grey) or detected as a case (orange, cumulative). Ribbons show movement between days; ribbons from grey back to colour are contacts seen again after a missed day (3,698 contacts at least once). Labels give the number and percentage seen on day 21. The eight health zones with the most contacts are shown; the rest are grouped as Other.



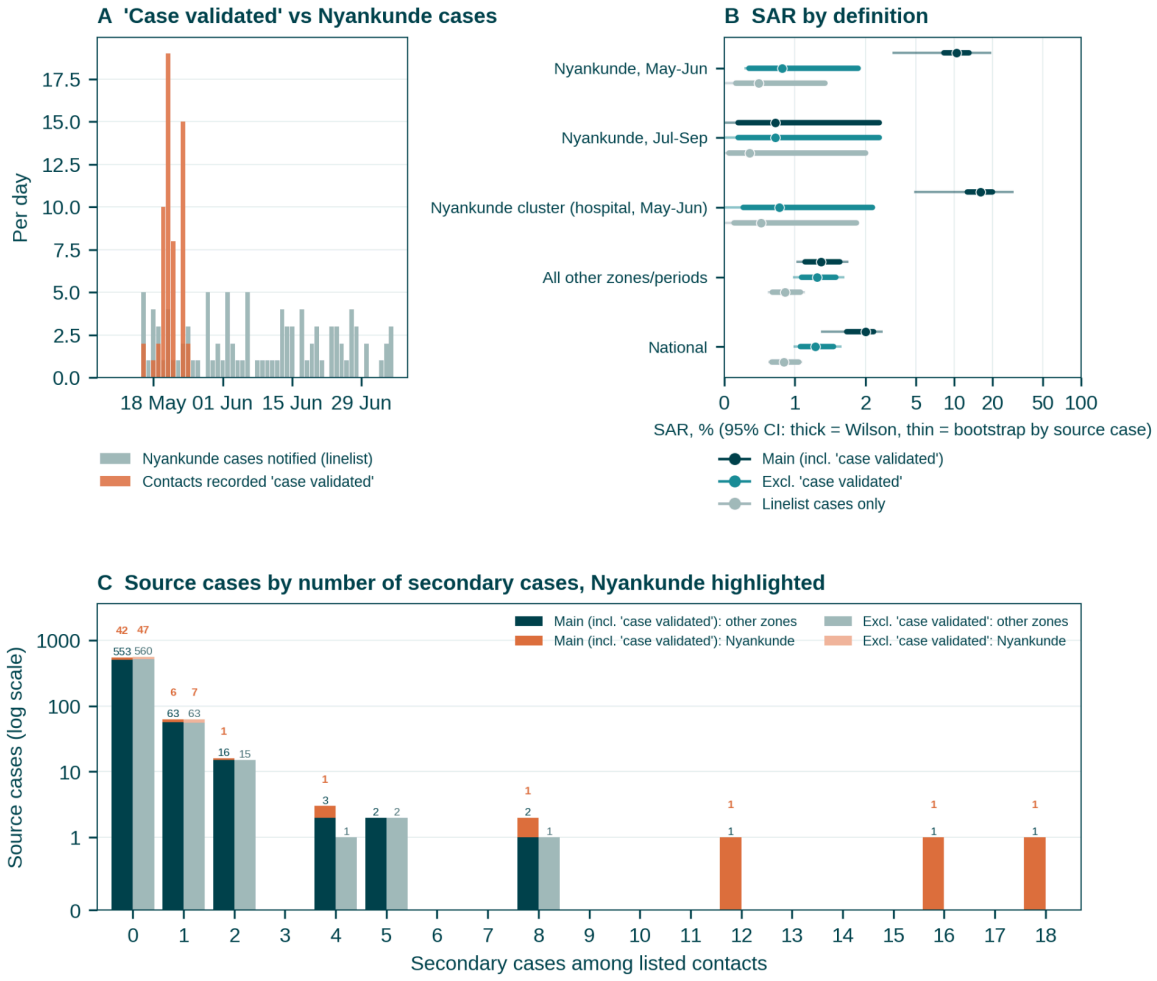
Data cut 9 Oct 2026; person-level after ID harmonisation.

Figure S3. (A) Days with a recorded follow-up result per contact (closed windows). (B) Same, by province of the source case.



Data cut 9 Oct 2026; person-level after ID harmonisation. Denominator: contacts of confirmed/probable source cases whose first follow-up was on or before 18 Sep 2026.

Figure S4. Sensitivity of secondary attack rate (SAR) to different inclusion criteria. Limiting the definition of secondary cases to only those who are found and confirmed in the linelist reduces SAR to 0.8%. Excluding ‘case validated’ or the Nyankunde May-June hospital cluster similarly reduces SAR.



Data cut 9 Oct 2026; person-level after ID harmonisation. Cluster = source in Nyankunde, hospital exposure, contact registered on or before 30 Jun 2026. C: SAR-eligible source cases; dark labels = all source cases, coral labels = of which Nyankunde.

Figure S5. Focused visualisations of the influence of the Nyankunde ‘case validated’ cluster. (A) Instances of ‘case validated’ in Nyankunde contact tracing entries (orange) plotted with the concurrent case trajectory according to the linelist. (B) Influence of eliminating ‘case validated’ (currently included in analysis in main) on SAR, both locally to Nyankunde during different time periods and nationally. (C) Source cases by number of secondary cases, with Nyankunde highlighted in orange. The three largest ‘super spreaders’, infecting 18, 16, and 12 secondary cases, were members of the Nyankunde ‘case validated’ hospital cluster.

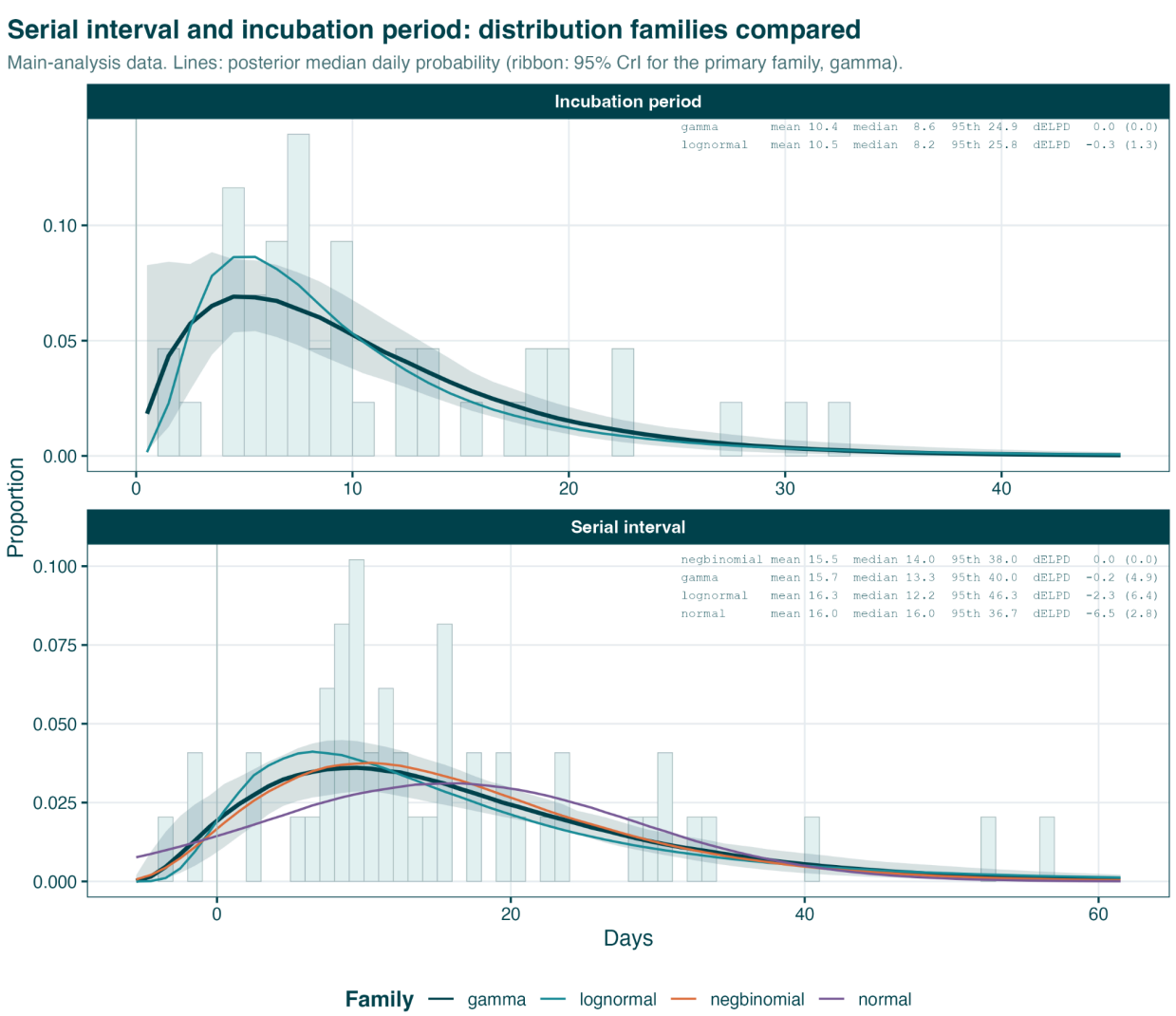


Figure S6. Fit of serial interval and incubation period data (using inclusion criteria for the main text) to different parametric distribution families. For both delay types, Weibull fits were implemented but failed to converge. For incubation period, negative binomial could not be used under the assumptions we make in the main text (exposure 'window' instead of static date) as it is fit using brms instead of EpiDist. Gamma distributions performed consistently under different iterations and sensitivity analyses for both delay types, so are used for the distribution fits shown in main and for the sensitivity analyses in S7 and S8. ELPD: Expected Log Predicted Density from Leave-One-Out Cross Validation, normalised to 0.0 for the best performing family and 'delta' (dELPD) for other families.

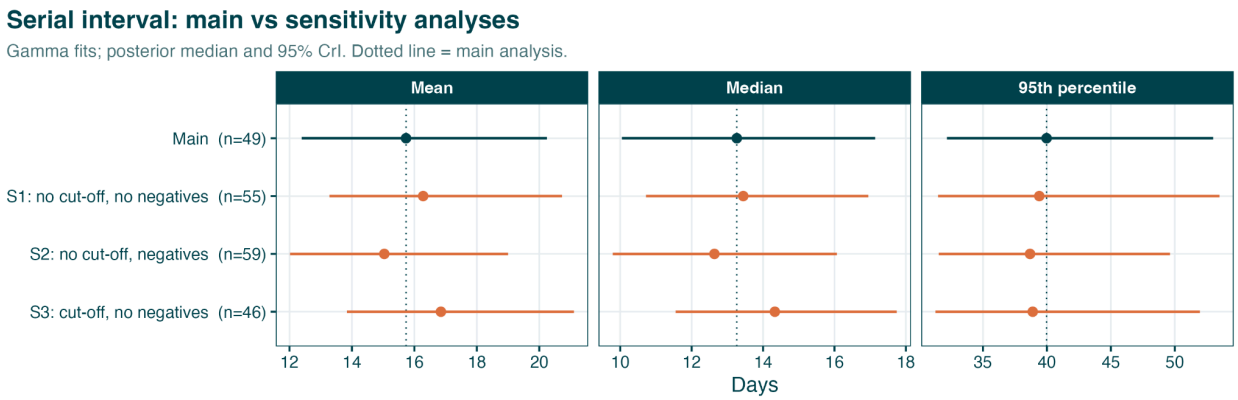


Figure S7. Sensitivity of serial interval to different inclusion criteria. S1: allowing pairs where infector onset was after 28th August, eliminating negative delays between infector and infectee onset. S2: allowing pairs where infector onset was after 28th August, but allowing up to -5 day delays between infector and infectee onset (as in main). S3: not allowing pairs where infector onset was after 28th August (as in main), but eliminating negative delays between infector and infectee onset.

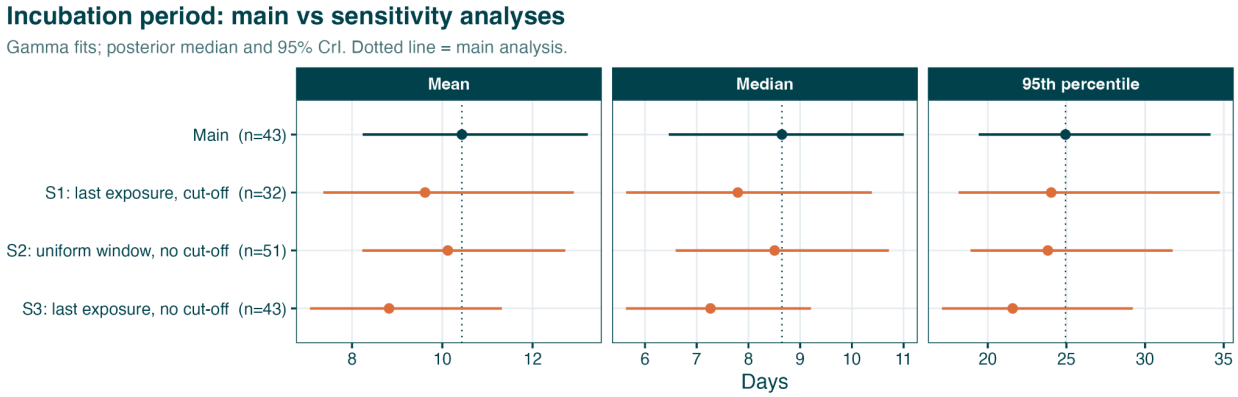
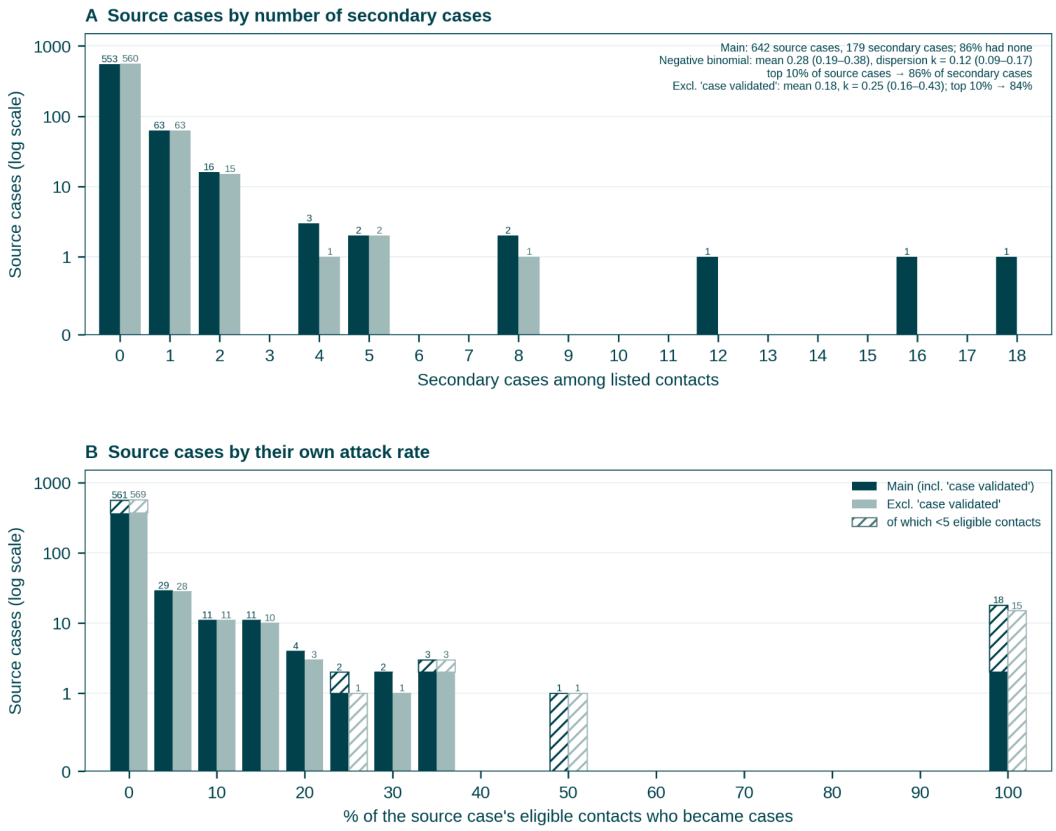


Figure S8. Sensitivity of incubation period to different inclusion criteria. S1: not allowing pairs where infector onset was after 28th August (as in main) & assuming transmission takes place on the last date of contact exposure to the source case S2: allowing pairs where infector onset was after 28th August & assuming a uniform distribution for transmission time between source case onset and the last date of contact exposure to the source case (as in main). S3: allowing pairs where infector onset was after 28th August & assuming transmission takes place on the last date of contact exposure to the source case.



Data cut 9 Oct 2026. Source cases = confirmed/probable cases with ≥1 contact whose follow-up window closed by 18 Sep 2026 (SAR denominator). Secondary cases found among listed contacts only (a lower bound). B: secondary cases ÷ eligible contacts of each source, in 5-point bins centred on 0, 5, …, 100%; hatched = sources with <5 eligible contacts. Log y-axis.

Figure S9. (A) Source cases by number of secondary cases, including and not-including the 'case validated' definition. Fitting a negative binomial to both datasets gave overdispersion parameter k values of 0.12 (0.09-0.17) and 0.25 (0.16-0.43) respectively. (B) Source cases plotted by the proportion of their contacts became positive (i.e. their individual attack rate), again including and not including the 'case validated' definition. Source cases with 100% individual attack rates were typically those with very few listed contacts (indicated by diagonal lines). This is with the exception of the inclusion of the 'case validated' definition, where some source cases with many contacts have reported an individual attack rate of 100%. This is likely an artefact of data entry (e.g., only positive contacts were entered onto the database).

References

1. Nash, R. K. *et al.* Ebola virus disease mathematical models and epidemiological parameters: a systematic review. *Lancet Infect. Dis.* **24**, e762–e773 (2024).

2. Estimate Epidemiological Delay Distributions with brms. https://epidist.epinowcast.org/.